DRUGS · NASAL PREPARATIONS

Intranasal corticosteroids

Intranasal corticosteroids are the single most effective treatment for allergic rhinitis and the foundation of medical treatment for chronic rhinosinusitis, with or without nasal polyps. They work on congestion, rhinorrhoea, sneezing and itch, and on the eye symptoms of allergic rhinitis. The modern molecules have very low systemic bioavailability, but they only work if they are used every day and sprayed correctly, and most treatment failures are failures of technique.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

INTRANASAL CORTICOSTEROID SPRAYS

	SUMMARY
Agents	Mometasone furoate; fluticasone propionate; fluticasone furoate; beclometasone dipropionate; budesonide; triamcinolone
How they work	Glucocorticoid receptor activation in the mucosa: fewer mast cells, eosinophils and T cells; less cytokine and chemokine release; less oedema and mucus
ENT uses	Allergic and non-allergic rhinitis; chronic rhinosinusitis with and without polyps; adjunct in acute post-viral and bacterial sinusitis; adenoidal hypertrophy and Eustachian tube dysfunction (modest evidence); after sinus surgery
Adult dose	Mometasone 2 sprays (50 micrograms each) into each nostril once daily, up to twice daily for polyps; fluticasone propionate 2 sprays each nostril once daily (maximum twice daily); fluticasone furoate 2 sprays each nostril once daily; beclometasone 2 sprays each nostril twice daily
Child dose	Mometasone 1 spray each nostril once daily from 3 years; fluticasone furoate 1 spray each nostril once daily from 6 years; fluticasone propionate 1 spray each nostril once daily from 4 years

	SUMMARY
Onset	Some effect within 12 hours; full effect after 1 to 2 weeks of regular use
Pregnancy	Budesonide and fluticasone have the most reassuring data; acceptable when needed

2 CHAPTER TWO How they work

Mechanism and handling.

2.1 Mechanism

They suppress both the early and late phases of the allergic response, though their effect on the early phase builds over days rather than minutes, which is why they work best taken regularly and started before the pollen season.

2.2 Systemic bioavailability

The molecules differ greatly in how much of the swallowed fraction survives first-pass metabolism: **mometasone and fluticasone** have under 1% to 2% systemic bioavailability, **budesonide** about a third and **beclometasone** almost half. Choose the low-bioavailability agents for children, for long-term use, and for patients also on inhaled steroids.

3 CHAPTER THREE Using them in ENT

Indications and technique.

3.1 Technique

Teach it at every visit:

- Blow the nose first; shake the bottle; prime a new bottle.

- **Head bent forward** (looking at the toes), not tipped back.
- **Use the opposite hand** (right hand for the left nostril) so the nozzle points **laterally**, away from the septum, toward the outer corner of the eye.
- Sniff gently, not hard (hard sniffing sends the dose to the throat).

3.2 Allergic rhinitis

First-line for moderate to severe or persistent symptoms, alone or with an antihistamine; more effective than oral antihistamines for congestion.

3.3 Chronic rhinosinusitis

Regular long-term use in every patient, in addition to saline irrigation; in polyposis, a higher dose, drops or rinses for better distribution (see steroid drops and irrigations), and short courses of oral steroids for exacerbations. After sinus surgery the opened cavities let sprays reach the sinuses better.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Epistaxis** and crusting (up to 1 in 10), mostly from spraying onto the septum; rarely **septal perforation**.
- Dryness, irritation, headache, taste and smell disturbance.
- Systemic effects are rare with mometasone and fluticasone; small reductions in growth velocity have been seen with beclometasone in children. Adrenal suppression is possible when high nasal doses are combined with inhaled and other steroids.

4.2 Interactions

Ritonavir and cobicistat (and other strong CYP3A4 inhibitors such as itraconazole) markedly raise fluticasone levels, causing Cushing syndrome and adrenal

suppression: use beclometasone or budesonide with care, or avoid.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism and **onset**, and why regular use matters.
- **Bioavailability** differences between agents and which to choose in children.
- Correct **technique** (head down, opposite hand, lateral aim).
- The **ritonavir** interaction.

5.2 Common pitfalls

COMMON PITFALLS

- Labelling a patient a treatment failure without checking technique and adherence.
- Spraying at the septum, producing bleeding and crusting.
- Using beclometasone long term in a small child.

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